Loris Bazzani

AI Research Leader & Adj. Professor | Multimodal & GenAI
PhD | 15+ Yrs in AI | Ex-Amazon

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http://scholar.google.com/citations?user=1cdNGL4AAAAJ&hl=en

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Current location: Berlin, Germany

Motivation

As an AI Research Leader with over 15 years bridging fundamental academic research and global-scale industry deployment, my mission is to architect transformational AI that solve critical real-world challenges and leave a lasting, positive footprint on society. Driven by a core belief in technology's ability to elevate human capability, I strive to foster innovation ecosystems rooted in shared knowledge, collaborative problem-solving, and creative exploration. Having spearheaded multi-year multimodal AI strategies and deployed production-grade deep learning pipelines, I am driven to build legacy-defining, cross-functional initiatives. My current focus lies in advancing adaptive multimodal intelligence for high-stakes applied domains, such as healthcare and digital services, where the synthesis of vision, language, and interactive systems fundamentally improves technological outcomes. I thrive at the intersection of scientific depth and operational execution, translating complex generative AI research into reusable assets and scalable solutions. I am looking for strategic leadership opportunities where I can unite academic institutions and industry partners to accelerate the time-to-deployment of foundational AI applications that genuinely benefit the global community.

Experience

Founding Research Leader - Stealth AI Start-up 2026-now

Adjunct Professor and Honorary Research Fellow - University of Verona (Part-time) 2025-now

- Driving multidisciplinary academic research on adaptive and collaborative multimodal intelligence for digital services and MedTech.
- Collaborating with research institutes and universities at global scale.
- Developing talent pathways by teaching Data Visualization to Master's level Data Science cohorts.

Principal Applied Scientist - Amazon 2016–2025

- Research Leadership & Strategy:
  - Defined and executed the AI strategy and tech agenda for Large Multimodal Models (LMMs) and Vision-Language Models (VLMs) across Amazon's consumer platforms.
  - Acted as the "Principal Investigator" for multi-year initiatives, driving complex resource planning and securing major internal investments (computing and headcount) by presenting research roadmaps to VP- and SVP-level leadership.
- Information Systems & Technical Impact:
  - Led high-impact core research and launched 8+ products across Prime Video, Alexa, and mobile/.com shopping, generating $100M+ in business impact. These included movies and TV series autotagging, image accessibility, live sports highlights, virtual try-on, interactive product recommendations, and shopping assistants used by millions of users worldwide.
  - Developed novel models and architectures for video understanding, VLMs, LMMs, and diffusion models, accelerating time-to-deployment by translating fundamental research into scalable, reusable assets.
- Multidisciplinary Team Scaling & Leadership:
  - Partnered with executive leadership to scale a multidisciplinary science organization from 4 to 15 PhD-level researchers, establishing rigorous peer-review and model-design standards.
  - Mentored 20+ Applied Scientists and served as the structural "glue" between applied research teams, engineering, and product organizations to successfully drive cross-functional projects to completion.
- Career Progression: rapidly promoted through three levels of seniority (Applied Scientist → Senior → Principal) based on sustained scientific impact, deployment of research and leadership in Multimodal AI.

Postdoc Researcher - Visual Learning Group - Dartmouth College 2014–2015

- Research on deep learning and computer vision, with production of joint publications and collaboration with industrial entities and international universities, on object recognition and video understanding; and basic teaching activities.

Co-founder and AI Scientist - VIBE 2012–2014

- Co-founded an AI spin-off from the Italian Institute of Technology, demonstrating an early drive to translate academic research into viable industry applications.
- Architected a privacy-compliant computer vision MVP to analyze consumer behavior and physical interactions.
- Led early-stage customer discovery, presenting technical demos to potential industry clients to validate product-market fit and commercial feasibility.

Postdoc Researcher - PAVIS Lab - Italian Institute of Technology 2012–2013

- Led research on computer vision and machine learning, participating in industrial projects, with a focus on social interaction analysis, person re-identification and object recognition using dynamic Bayesian nets and Bayesian non-parametric models, and kernel-based methods.

Visiting Student - Laboratory for Computational Intelligence - University of British Columbia 2010

- Explored deep generative neural models, including restricted Boltzmann machines, attentional models, and sequential Monte Carlo methods, laying foundational knowledge that informed later work in deep learning and generative AI.

Education

Ph.D. in Computer Science Computer - University of Verona 2009–2012

- PhD Thesis: Beyond multi-target tracking: statistical pattern analysis of people and groups
- Supervisors: V. Murino and M. Cristani. Reviewers: A. Del Bimbo and R. T. Collins
- Task leader for some work packages of the Europen project FP7-217899 SAMURAI

M.S. in Intelligent and Multimedia Systems - University of Verona 2006–2008

Thesis topics: particle filtering and multi-person tracking. Summa cum laude and top of the class

B.S. in Information Technology - University of Verona 2003–2006

Thesis topics: machine learning for medical imaging

Research & Publications

Multimodal Learning with Interaction/User Feedback

- N. Subedi, L. Bazzani, Z. Al-Halah. Interactive Episodic Memory with User Feedback, CVPR 2026.
- E. Zorzi, F. Taioli, Y. Wang, M. Cristani, A. Farinelli, A. Castellini, L. Bazzani. Benchmarking Interaction, Beyond Policy: a Reproducible Benchmark for Collaborative Instance Object Navigation, Arxiv 2026.
- Z. Wang, S. Ramasinghe, C. Xu, J. Monteil, L. Bazzani, T. Ajanthan. Learning Visual Hierarchies in Hyperbolic Space for Image Retrieval, ICCV 2025.
- M. Guillaumin, M. Jaritz, S. Sternig, L. Bazzani. UniCoRN: Unified Commented Retrieval Network with LMMs, Arxiv 2025.
- V. Pippi, M. Guillaumin, S. Casciarelli, R. Cucchiara, M. Jaritz, L. Bazzani. ToFu: Visual Tokens Reduction via Fusion for Multi-modal, Multi-patch, Multi-image Task, Arxiv 2025.
- A. Salvador, E. Gundogdu, L. Bazzani, M. Donoser. Revamping Cross-Modal Recipe Retrieval with Hierarchical Transformers and Self-supervised Learning, CVPR 2021.
- A. D'Innocente, N. Garg, Y. Zhang, L. Bazzani, M. Donoser. Localized Triplet Loss for Fine-Grained Fashion Image Retrieval, CVPR Workshops 2021.
- Y. Hou, E. Vig, M. Donoser, L. Bazzani. Learning Attribute-driven Disentangled Representations for Interactive Fashion Retrieval, ICCV 2021.
- Y. Chen, and L. Bazzani. Learning Joint Visual Semantic Matching Embeddings for Language-guided Retrieval, ECCV 2020.

- Y. Chen, S. Gong, and L. Bazzani. Image Search with Text Feedback by Visiolinguistic Attention Learning, CVPR 2020.

Multimodal Federated Learning

- A. Chhetri, B. Niroula, E. Vasquez, Y. R. Shrestha, P. K. Gyawali, L. Bazzani, B. Bhattarai. ProMoE-FL: Prototype-conditioned Mixture of Experts for Multimodal Federated Learning with Missing Modalities, MICCAI 2026.
- A. Chhetri, B. Niroula, P. Shrestha, Y. R. Shrestha, L. A. Anderson, P. K. Gyawali, L. Bazzani, B. Bhattarai. Med-MMFL: A Multimodal Federated Learning Benchmark in Healthcare, Arxiv 2026.

Multimodal Generative Models

- Z. Liu, D. Talon, F. Girella, Z. Ruan, M. Mondo, L. Bazzani, Y. Wang, M. Cristani. Multi-Level Conditioning by Pairing Localized Text and Sketch for Fashion Image Generation, Arxiv 2026.
- A. Cao, M. Jaritz, M. Guillaumin, R. de Charette, L. Bazzani. LatteCLIP: Unsupervised CLIP Fine-Tuning via LMM-Synthetic Texts, WACV 2025.
- X. Yang, Y. Zuo, S. Ramasinghe, L. Bazzani, G. Avraham, A. van den Hengel. ViewFusion: Towards Multi-View Consistency via Interpolated Denoising, CVPR 2024.
- R. Bodur, E. Gundogdu, B. Bhattarai, T-K Kim, M. Donoser, L. Bazzani. iEdit: Localised Text-guided Image Editing with Weak Supervision, CVPR Workshops 2024.
- L. Bazzani, T. Domhan, and F. Hieber. Image Captioning as Neural Machine Translation Task in SOCKEYE, Arxiv 2018.

Video Understanding

- M. Xu, E. Gundogdu, M. Lapin, B. Ghanem, M. Donoser, L. Bazzani. Contrastive Language-Action Pre-training for Temporal Localization, Arxiv 2022.
- L. Bazzani, H. Larochelle, L. Torresani. Recurrent Mixture Density Network for Spatiotemporal Visual Attention, ICLR 2017.
- V. Murino, S. Gong, C. C. Loy and L. Bazzani, Image and Video Understanding in Big Data, Special Issue in CVIU 2017.
- L. Bazzani, M. Zanotto, M. Cristani, V. Murino, Joint individual-group modeling for tracking, PAMI 2015.
- L. Bazzani, V. Murino, and M. Cristani, Decentralized particle filter for joint individual-group tracking, CVPR 2012.
- M. Denil, L. Bazzani, H. Larochelle, and N. de Freitas, Learning where to attend with deep architectures for image tracking, Neural Computation 2012.
- L. Bazzani, N. de Freitas, H. Larochelle, V. Murino, J. Ting, Learning attentional policies for object tracking and recognition in video with deep networks, ICML 2011.
- L. Bazzani, N. de Freitas, J. Ting, Learning attentional mechanisms for simultaneous object tracking and recognition with deep networks, NeurIPS Workshop 2010.
- L. Bazzani, M. Cristani, and V. Murino, Collaborative particles filters for group tracking, ICIP 2010.

Structured Methods for Images

- L. Bazzani, A. Bergamo, D. Anguelov, L. Torresani. Self-taught object localization with deep networks, WACV 2016.
- M. San Biagio, H. Q. Minh, L. Bazzani, V. Murino. Approximate Log-Hilbert-Schmidt distances between covariance operators for image classification, CVPR 2016.
- H. Q. Minh, L. Bazzani, V. Murino, A unifying framework in vector-valued reproducing kernel Hilbert spaces for manifold regularization and co-regularized multi-view learning, JMLR 2016.
- H. Q. Minh, M. San Biagio, L. Bazzani, and V. Murino, Kernel Methods on Approximate Infinite-Dimensional Covariance Operators for Image Classification, Arxiv 2016.
- M. San Biagio, L. Bazzani, M. Cristani, V. Murino, Weighted bag of visual words for object recognition, ICIP 2014.
- H. Q. Minh, L. Bazzani, V. Murino, A unifying framework for vector-valued manifold regularization and multi-view learning, ICML 2013.

Behavioral and Interaction Analysis in Videos

- S. Vascon, and L. Bazzani, Group Detection and Tracking using Sociological Features, Group and Crowd Behavior for Computer Vision 2017.
- L. Bazzani, D. Tosato, M. Cristani, M. Farenzena, G. Paggetti, G. Menegaz, V. Murino, Social interactions by visual focus of attention in a three-dimensional environment, Expert Systems 2013.
- M. Zanotto, L. Bazzani, M. Cristani, and V. Murino, Online bayesian non-parametrics for social group detection, BMVC 2012.
- L. Bazzani, M. Cristani, G. Pagetti, D. Tosato, G. Menegaz, V. Murino, Analyzing groups: a social signaling perspective, Video Analytics for Business Intelligence 2012.
- M. Cristani, L. Bazzani, G. Paggetti, A. Fossati, A. Del Bue, D. Tosato, G. Menegaz, V. Murino, Social interaction discovery by statistical analysis of F-formations, BMVC 2011.
- M. Cristani, G. Paggetti, A. Vinciarelli, L. Bazzani, G. Menegaz, V. Murino, Towards computational proxemics: Inferring social relations from interpersonal distances, SocialCom 2011.

Cross-Camera Identification

- L. Bazzani, M. Cristani, and V. Murino. SDALF: Modeling human appearance with symmetry-driven accumulation of local features, Person Re-identification 2014.
- D. Figueira, L. Bazzani, H.Q. Minh, M. Cristani, A. Bernardino, and V. Murino, Semi-supervised multi-feature learning for person re-identification, AVSS 2013.
- P. Salvagnini, L. Bazzani, M. Cristani, and V. Murino, Person re-identification with a PTZ camera: an introductory study, ICIP 2013.
- L. Bazzani, M. Cristani, and V. Murino. Symmetry-driven accumulation of local features for human characterization and re-identification, CVIU 2013.
- B. I. Barbosa, M. Cristani, A. Del Bue, L. Bazzani, and V. Murino, Re-identification with RGB-D sensors, ECCV Workshops 2012.
- L. Bazzani, M. Cristani, A. Perina, V. Murino, Multiple-shot person re-identification by chromatic and epitomic analyses, PRL 2012.
- D. S. Cheng, M. Cristani, M. Stoppa, L. Bazzani, V. Murino, Custom pictorial structures for re-identification, BMVC 2011.
- L. Bazzani, M. Cristani, A. Perina, M. Farenzena, and V. Murino, Multiple-shot person re-identification by HPE signature, ICPR 2010.
- M. Farenzena, L. Bazzani, A. Perina, V. Murino, and M. Cristani Person re-identification by symmetry-driven accumulation of local features, CVPR 2010.

Blue color = top-tier computer vision/machine learning conferences or journals.

Patents

Secured US Patents directly linked to deployed, revenue-generating multimodal AI and recommendation systems, demonstrating a consistent ability to translate foundational research into commercialized intellectual property.

- US Patent 11,720,942: Interactive retrieval using visual semantic matching, 2023.
- US Patent 11,809,520: Localized visual similarity, 2023.
- US Patent 11,829,445: Attribute-based interactive product recommendations, 2023.
- US Patent 11,416,910: Visual blending of content, 2022.
- US Patent 11,361,212: Machine learning system to score alt-text in image data, 2022.
- US Patent 10,643,074: Automated video ratings, 2020.

Technical & Infrastructure

- AI Applied Prototyping: Python, PyTorch, HuggingFace, generative AI coding assistants, Git, Bash, and LaTeX. Legacy experience integrating MATLAB, Lua/Torch7, and JavaScript.
- Cloud and Scale: Deep expertise in architecting and orchestrating AWS services (EC2, Batch, Lambdas) and managing large-scale GPU computing clusters for model training and deployment.
- Innovation: Passionate about rapid technical problem-solving, having actively built prototypes and led technical teams in annual internal AI hackathons.

Memberships, Awards & Grants

- **Member of European Laboratory for Learning and Intelligent Systems (ELLIS)**: selected European scientists that publish regularly in top-tier conferences in AI, 2025.
- **Outstanding Reviewers @ CVPR**, recognition that reflects not only the quality and depth of reviews but also your commitment to fairness, clarity, and timely feedback, 2025.
- **Amazon Accessibility Innovation Award**, internal award given to individuals who created experiences to improve accessibility, 2021.
- **Amazon Robotics Challenge Award**, internal award given to the team who won the robotics challenge, 2018.
- **Finalist at SmartCup Liguria 2013 with VIBE**, 5 startups were selected at the regional level to enter this stage where they were presented to a panel of expert judges, 2013.
- **IBM Best Student Paper Award**, track: Computer Vision at International Conference on Patter Recognition ICPR, 2010.
- **Scholarship** from University of Verona that supported my Ph.D. from Jan. 2009 to Dec. 2011.
- **Scholarship** from EU-Project FP7 SAMURAI, grant FP7-SEC-2007-01 No. 217899, that contributed to support my Ph.D. from Jan. 2009 to Dec. 2011.
- **Travel grant** from University of British Columbia to attend Neural Information Processing Systems, 2010.
- **Travel grant** from International Machine Learning Society to attend the International Conference on Machine Learning, 2011.

AI Community Efforts

- Area Chair for NeurIPS 2026 and Program Chair for Annual Nepal AI School (ANAIS) 2026
- Co-organizer of the workshop on Human-AI Co-Creation @ ECCV 2026
- Co-organizer of the workshop on Computer Vision for Fashion, Art, and Design @ CVPR/ICCV 2021-2025
- Co-organizer of the workshop on 3D Vision and Modeling Challenges in eCommerce @ ICCV/ECCV 2023-2024
- Co-organizer and Area/Program Chair of 8 AI conferences and shopping science summits internal at Amazon
- Reviewer for CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, IEEE TPAMI, IJCV, IEEE Transactions on Multimedia, IEEE Transactions on Image Processing

Italian Scientific Qualifications

ASN Habilitation for Associate Professor in Computer Science (01B1) 2025-now
Certification given by the Italian Minister of Education reviewed by a national committee.

ASN Habilitation for Associate Professor in Information Processing Systems (09H1) 2026-now
Certification given by the Italian Minister of Education reviewed by a national committee.

Academic & Industry Ecosystem Engagement

Invited Speaker in Multiple Venues

- Delivered 50+ technical presentations across universities, public conferences, and internal Amazon events to audiences of up to 1,000 people.

- Translated complex AI research into accessible, high-impact narratives, successfully bridging the communication gap between deep academic theory and practical industry application.
- Selected recent invited talks: "Multimodal Retrieval: From Visual Hierarchies to Interactive and Generative Models" presented at IIT, University of Trento/FBK and Politecnico di Torino in 2025; "Image Generation, Editing and Retrieval with Interactive Elements" presented at Meta AI and the Amazon StatML workshop.

Academic Teaching

- Annual Nepal AI School & AI Summer School at the Zhejiang Normal University (Upcoming, 2026): invited speaker, fostering international AI talent and cross-border research collaborations.
- Data Visualization at University of Verona (2025): taught the course which is 48 hours and 6 CFUs.
- Machine Learning at Dartmouth College (2015): assistant professor for 10 hours of the undergrad course.
- Introduction to Machine Learning and Computer Vision at the Italian Institute of Technology (2013): co-organized and taught the course for PhD students.
- Person Re-identification at University of Modena and Reggio Emilia (2011): co-organized and taught the course for PhD students at the Short Spring School in Surveillance.